VU TRONG CHAU

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WORK EXPERIENCE

AI/ML Engineer Mar 2026 – Present
TELUS Digital *Remote*

- Evaluated and optimized large language models (LLMs), improving alignment between natural language intent and executable outputs
- Designed RLHF and supervised fine-tuning (SFT) workflows to enhance model accuracy, safety, and multi-step reasoning
- Validated AI-generated Python and JavaScript code across frameworks (React, Django), ensuring production-level quality
- Performed security analysis on model outputs, identifying vulnerabilities such as SQL injection and unsafe operations
- Built human-in-the-loop (HITL) evaluation pipelines and monitored model performance using drift detection and benchmarking techniques

AI Software Development Intern Aug 2025 – May 2026
SitusAMC *Remote*

- Designed end-to-end ML pipelines for loan risk prediction using 890K+ financial records, enabling scalable and reliable model deployment
- Engineered features from borrower and macroeconomic datasets (FRED, BLS, BEA) to improve model performance and interpretability
- Developed and optimized ML models (Logistic Regression, Random Forest, Gradient Boosting) using Scikit-learn
- Built scalable ETL pipelines for data ingestion and transformation, supporting consistent data flow across environments
- Developed API-ready outputs and integrated models into production-style workflows, enabling real-time analytics and decision support

Advanced Process Engineering (APE) Engineer Aug 2021 – Dec 2023
Techtronic Industries (TTI) *Ho Chi Minh, Vietnam*

- Led data-driven optimization of high-volume manufacturing lines using regression analysis, DOE, and multivariate statistical modeling
- Built structured datasets from IoT sensor logs and quality metrics for large-scale root cause analysis
- Developed Python-based ETL pipelines and KPI dashboards to automate reporting and surface actionable production insights
- Reduced defect rates and improved process yield through statistical analysis and iterative A/B testing of process parameters

PROJECTS

Healthcare Chatbot | *Python, Flask, LangChain, FAISS, PyTorch, FastAPI, Docker, CI/CD* Sep 2025 – May 2026

- Designed and deployed an LLM-powered healthcare assistant using Retrieval-Augmented Generation over 250,000+ medical records
- Implemented FAISS vector indexing, optimized prompt engineering, improving retrieval accuracy by 25% and enabling low-latency (<500ms)
- Architected FastAPI microservices with Docker-based CI/CD pipelines, reducing deployment time by 40% and supporting real-time inference

Threat Detection using Machine Learning | *Python, Scikit-learn, XGBoost, Flask, Pandas, NumPy* Jan 2025 – July 2025

- Developed ML models (Logistic Regression, XGBoost, Random Forest, Naive Bayes), achieving 92% accuracy on 72,000+ labeled texts
- Engineered semantic, syntactic, and sentiment features, increasing classifier performance by 15%
- Built a real-time moderation interface using Flask, generating severity scores and reducing review time by 40%

Global Population Prediction | *Python, Pandas, NumPy, D3.js, ETL Pipelines, Data Visualization, Flask* Jan 2025 – May 2025

- Forecasted population trends (1960–2023) for 200+ countries using ML models, achieving <2% MAE
- Built interactive dashboards with choropleth maps and time-series visualizations to support research insights
- Automated scalable ETL pipelines reducing manual data updates by 60%

Sleep Quality Prediction | *Python, Scikit-learn, TensorFlow, CNNs, RNNs, Flask* Sep 2024 – Dec 2024

- Built ML and deep learning models (CNNs, RNNs, Random Forest) to predict sleep quality from 70,000+ lifestyle and biometric records
- Improved precision by 20% over baseline models through feature engineering and model tuning
- Developed a real-time Flask interface providing personalized recommendations

TECHNICAL SKILLS

Programming Languages: Python, SQL, C/C++, JavaScript
AI & Machine Learning: LangChain, FAISS, Hugging Face, Scikit-learn, TensorFlow, PyTorch, CNNs, RNNs, XGBoost
LLM & GenAI: RAG pipelines, Prompt Engineering, RLHF dataset creation, LLM evaluation & benchmarking
MLOps & Data: FastAPI, Docker, CI/CD, ETL Pipelines, Pandas, NumPy, Tableau, Streamlit, AWS, GCP

EDUCATION

Troy University Troy, Alabama
Master of Science in Computer Science - Artificial Intelligence | GPA: 3.5 *July 2025*

University of Sunderland Sunderland, UK
Bachelor of Engineering in Electronic and Electrical Engineering | UK 2:1 Honours (3.5 GPA equivalent) *July 2021*